D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

VISUALIZING HOUSING MARKETS TRENDS: AN ANALYSIS OF SALE PRICES AND FEATURES USING TABLEAU

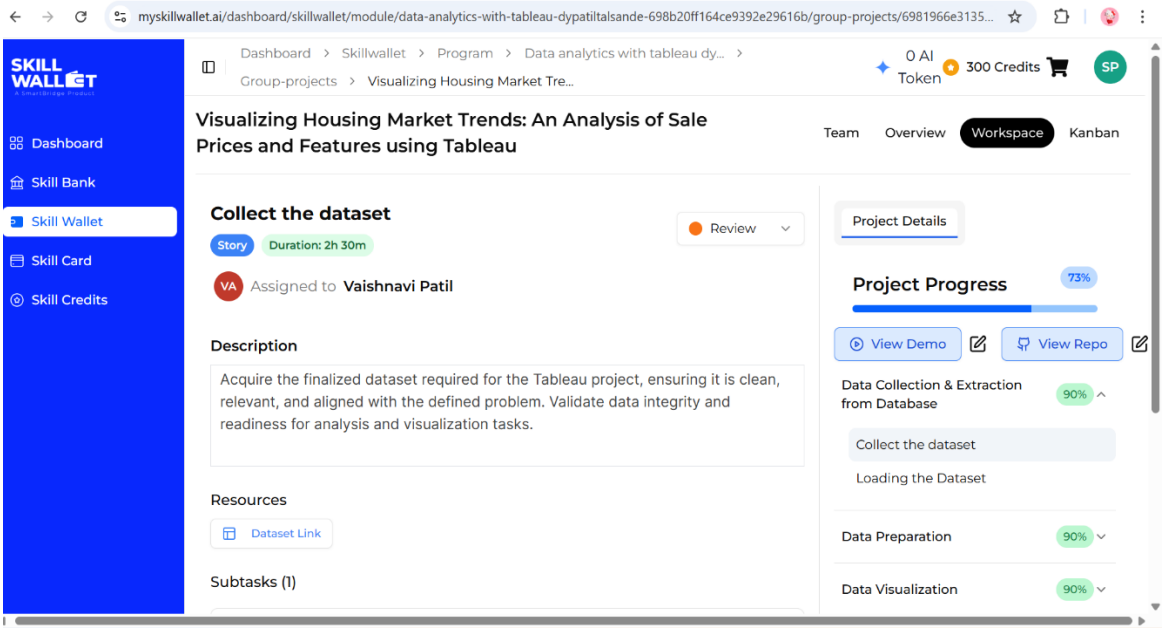
TEAM MEMBERS

1) GITANJALI PATIL
2) SAKSHI PATIL
3) VAISHNAVI PATIL
4) MITESH DESAI

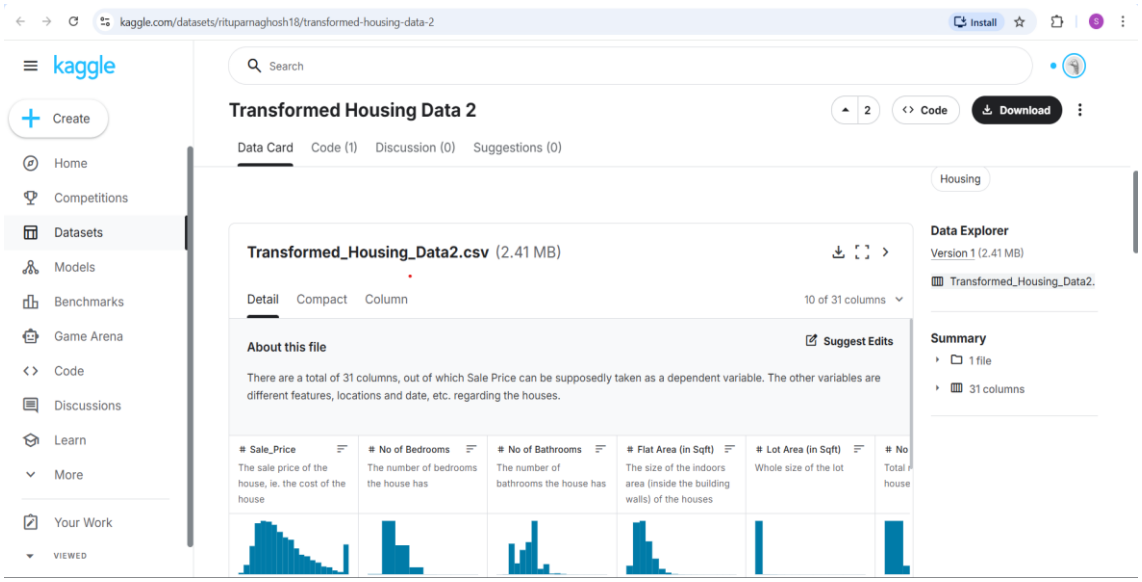
1) Data Collection & Extraction from Database:

❖ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the Kaggle Download the Dataset **Transformed_Housing_Data2.csv** by clicking on **Download** button,



❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File"
On the Start Page, under Connect, you will see options like:

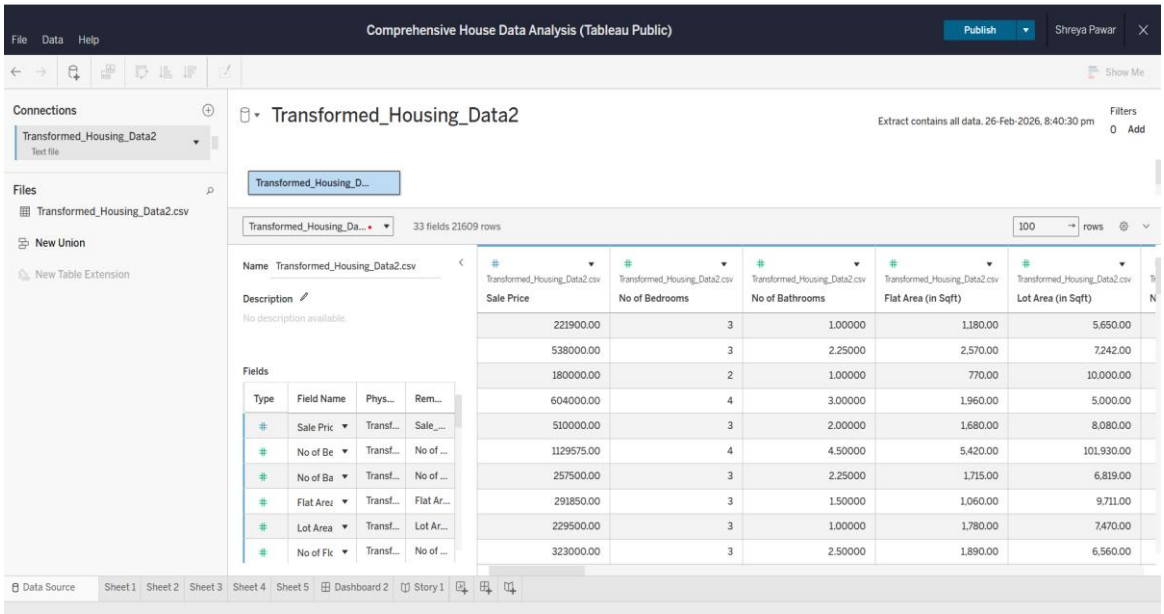
Microsoft Excel (for .xlsx)
Text File (for .csv)
Click the correct option.

Step 2: Select Your File

Browse your computer.
Select your dataset file.
Click Open.

Step 3: Data Source Page

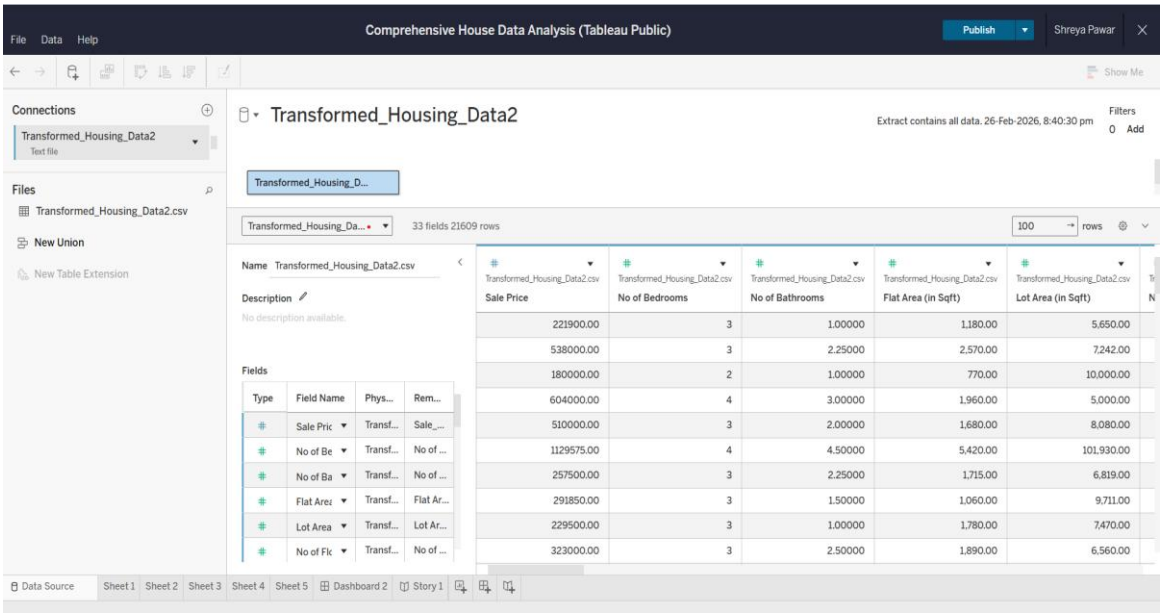
Your sheet names will appear on the left side.
Drag the sheet to the centre area.
Tableau will show a preview of your data.



2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



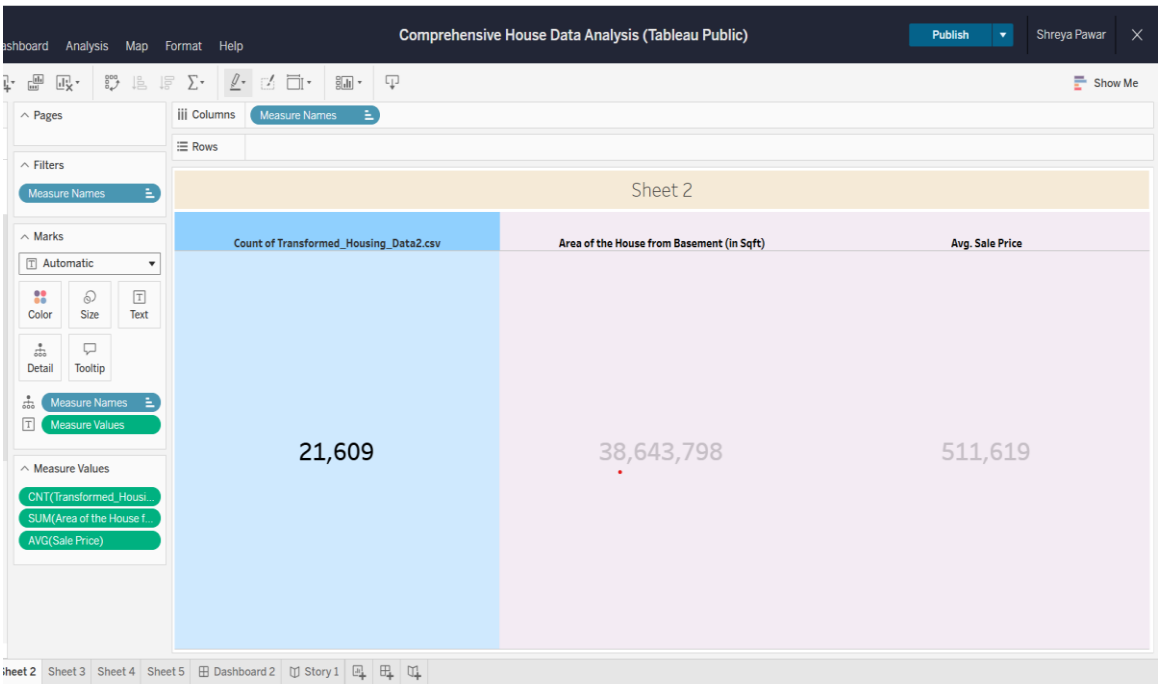
3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

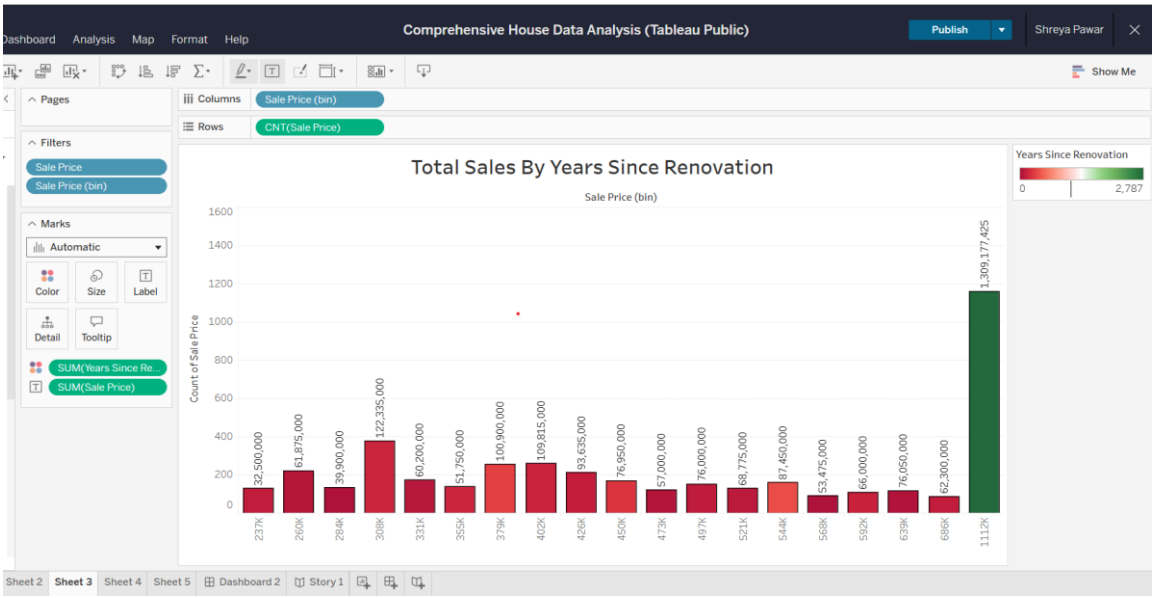
❖ No of Unique Visualizations:

1) Count of Transformed_Housing_Data2.cvs

a) Drag and Drop the Measure Name and Measure Values.
b) Add only-
   - Count of Transformed_Housing_Data2.cvs
   - Area of the House from Basement (in Sqft)
   - Avg.SalePrice
c) Drag and Drop Measure Name at Column section.

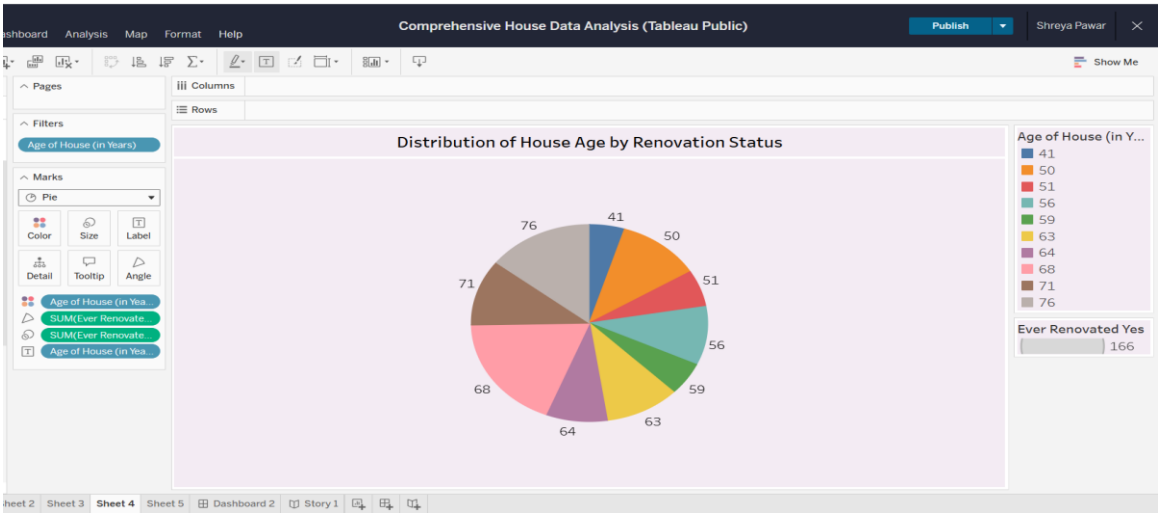


2) **Total Sales by Years Since Renovation**
   a) Convert the Sales Price into Dimension by right clicking.
   b) Add Filters into Sales Price and Sales Price(bin) to display Top 25 records.
   c) Drag and Drop Sales Price(bin) for grouping Sales Price into column section, and also CNT(Sales price) into rows.
   d) Drag and Drop SUM(Year Since Renovation) into color and SUM(Sale Price) into Lable.

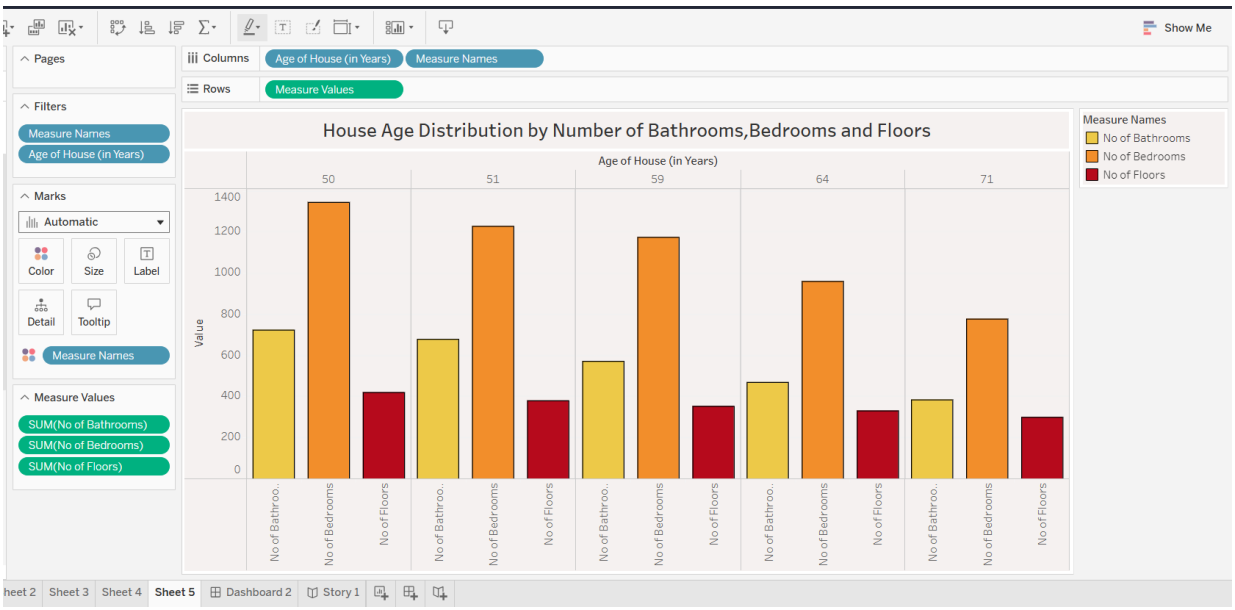


3) **Distribution of House Age by Renovation Status**
   a) Convert Age of House(in Years) to Dimension.
   b) Filter Age of House(in Years) into Top 10 records.

c) Add Age of House(in Years) into color.
d) Add SUM(Ever Renovated yes) into angle.
e) Add Sum(Ever Renovated yes) into size.
f) Add Age of House(in Years) in to label.



4) **House Age Distribution by Number of Bathrooms, Bedrooms and Floors**
   a) Drag and Drop Measure Name and Age of House(in Years).
   b) Filter Age of House(in Years) into Top 5 records.
   c) Drag and Drop Measure Values-
      - SUM(No of Bathroom)
      - SUM(No of Bedroom)
      - SUM(No of Floors)
   d) Drag and Drop Measure Names into Color.
   e) Add Age of House(in Year) and Measure Name into column section, and Measure Values into rows.



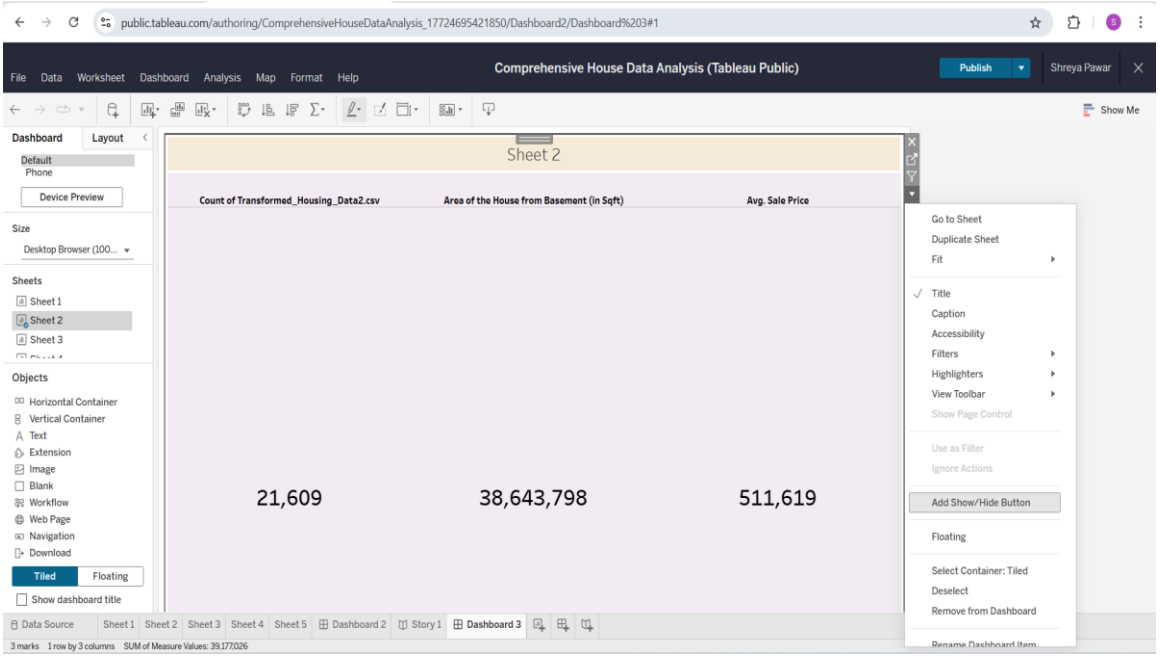
4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

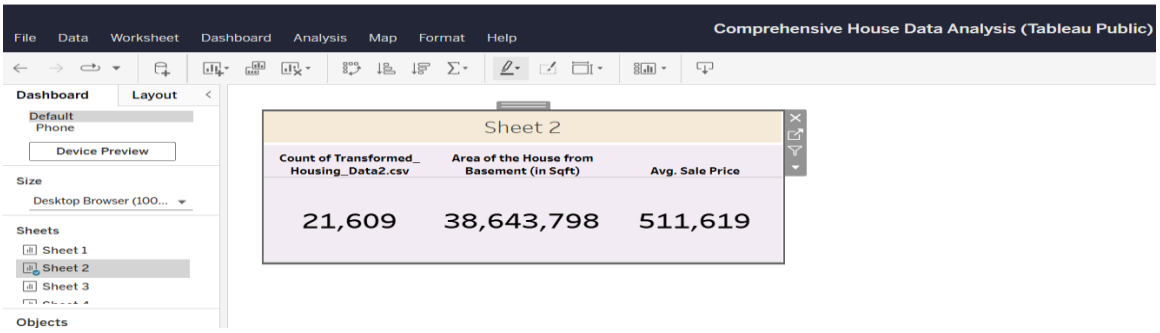
❖ Responsive and Design of Dashboard

a) Click On **New Dashboard** option.

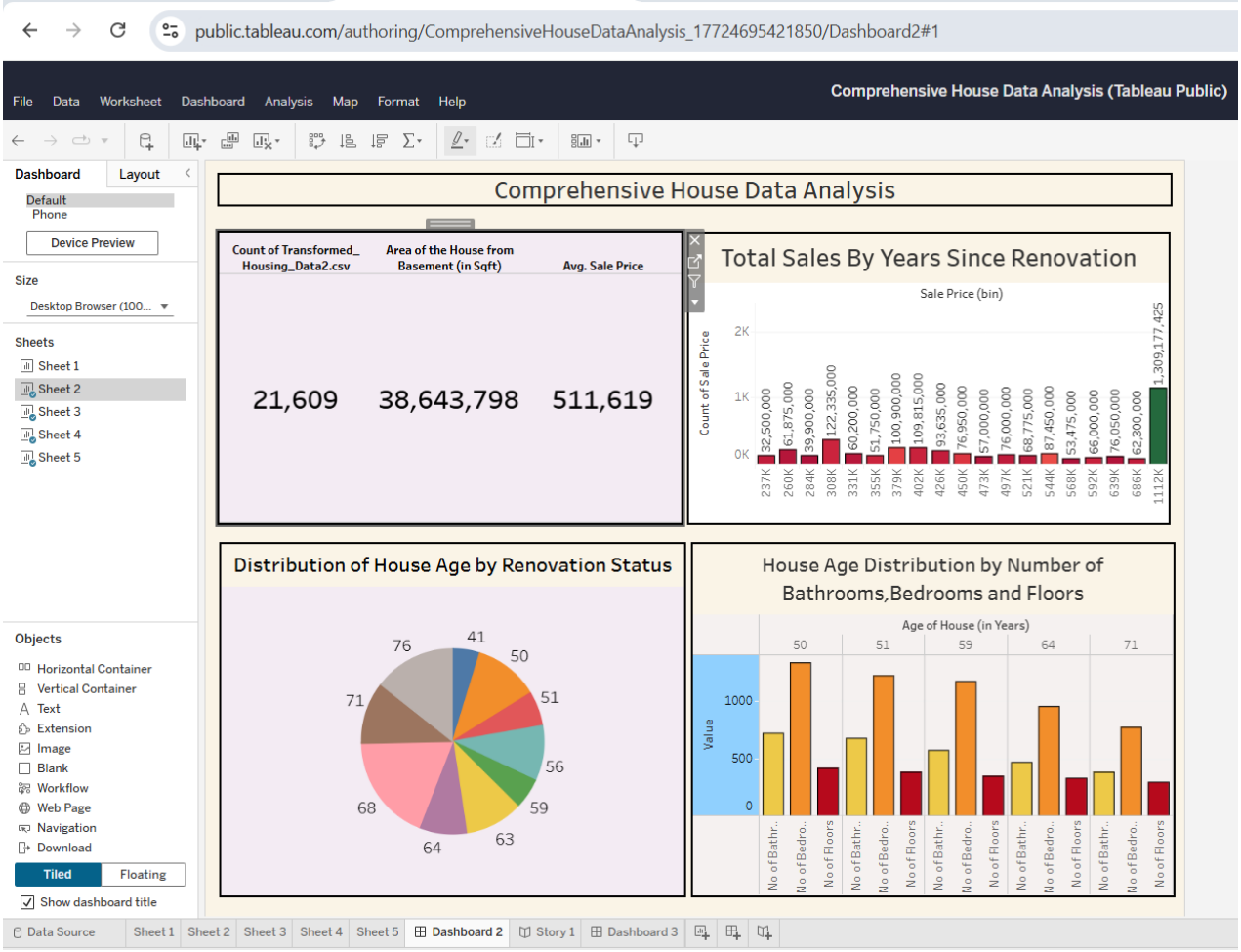
b) Drag and Drop Sheets from the data panel into centre white screen and in the right side of sheet there is Downward arrow so click on that and choose **Floating** option.



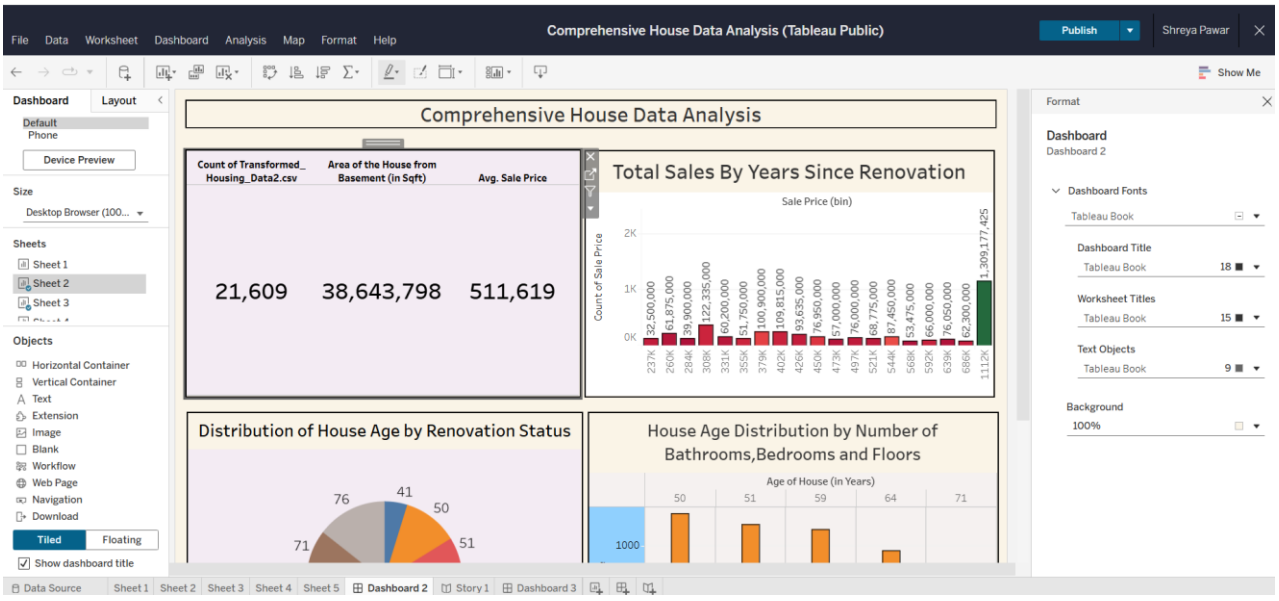
c) After selecting Floating option it's looks like.



d) Drag and Drop the Required Sheets for dashboard there I added 4 sheets.



e) Add a Format to an Dashboard by clicking at Format option at the Top and add background color to dashboard and boarder to the sheets.

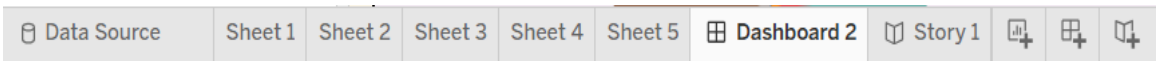


5) Storyboard

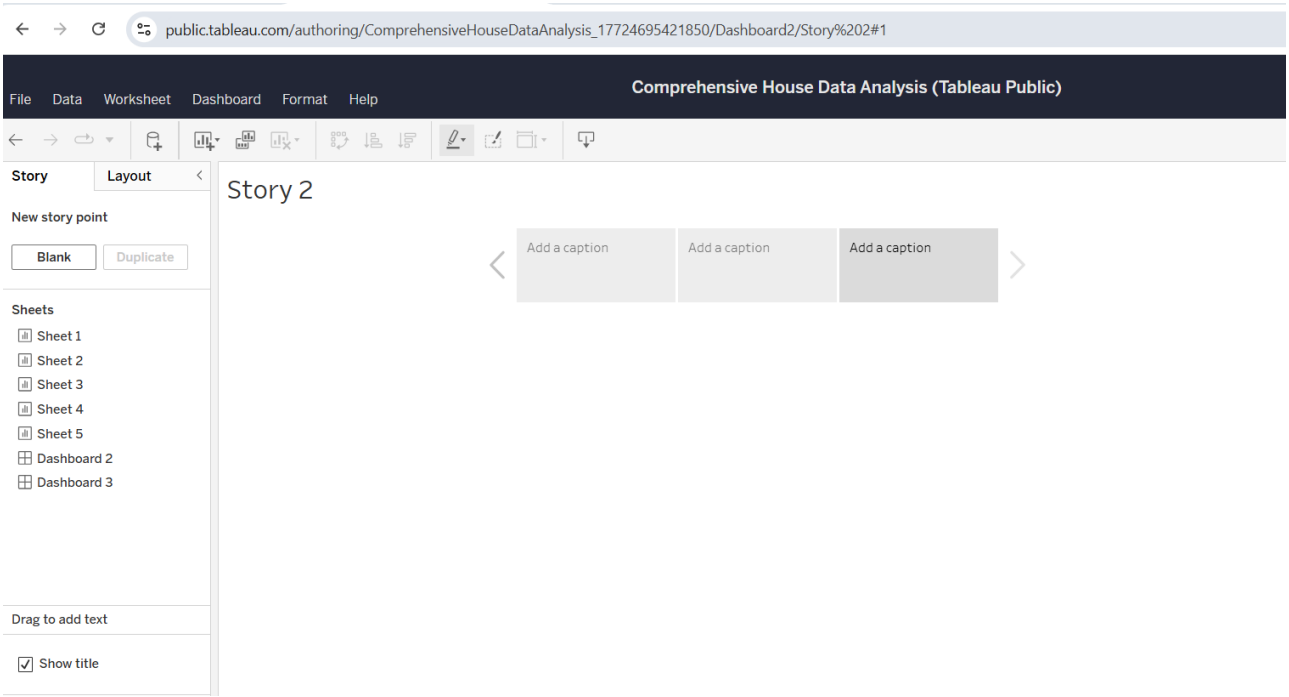
A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

❖ Design a Storyboard

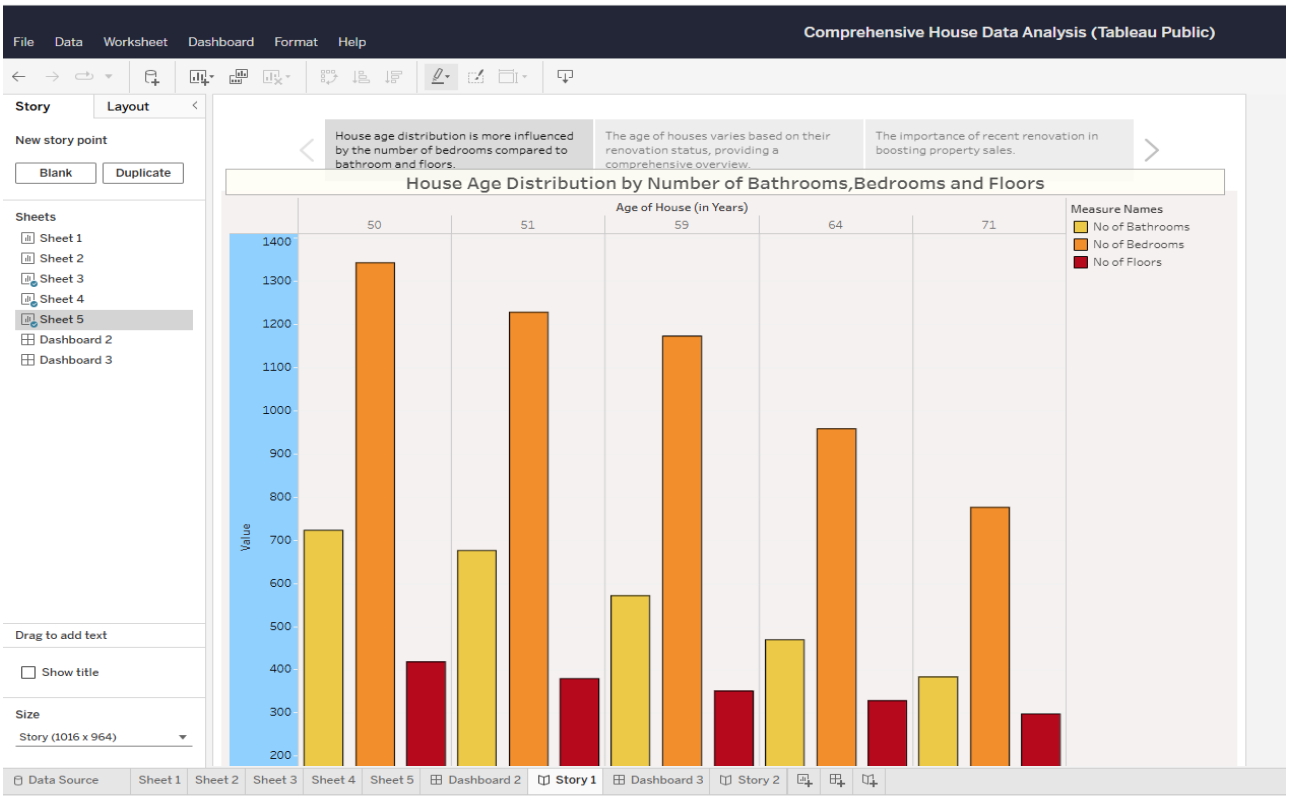
a) Click On **New Story** option.

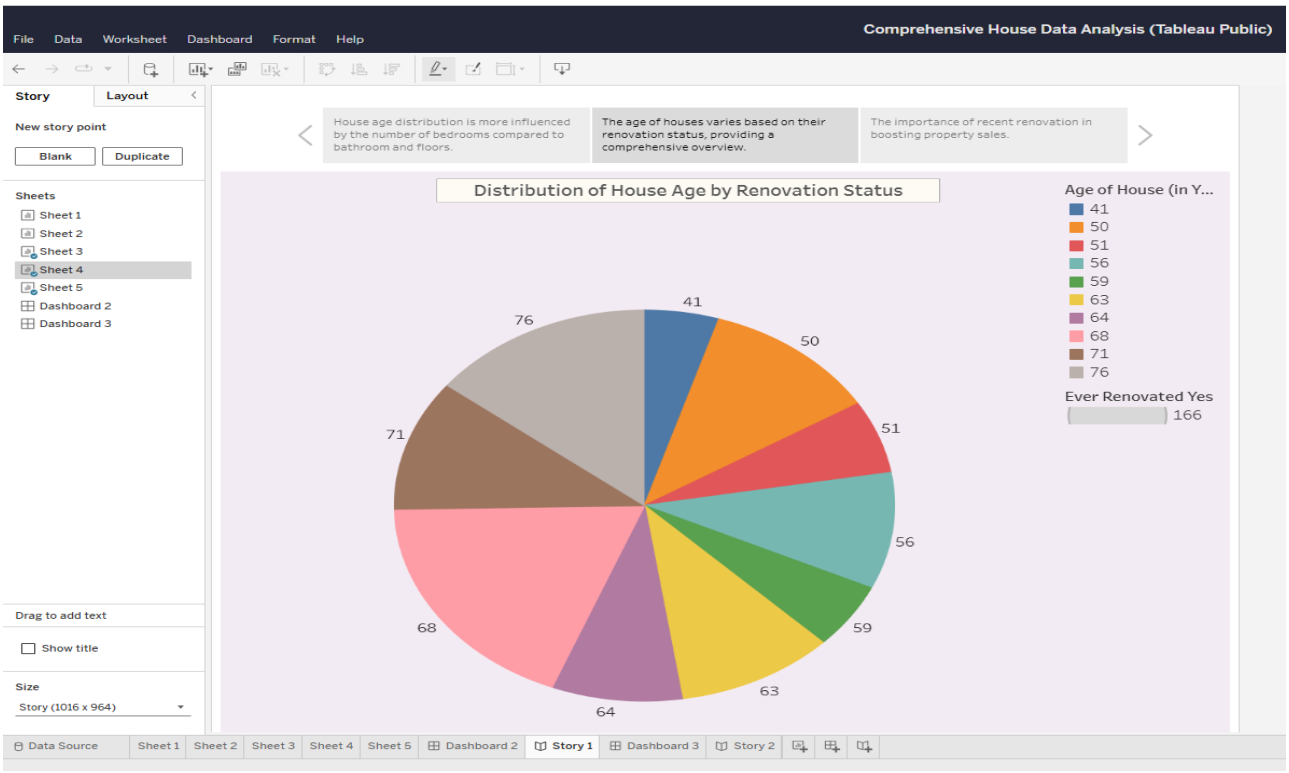


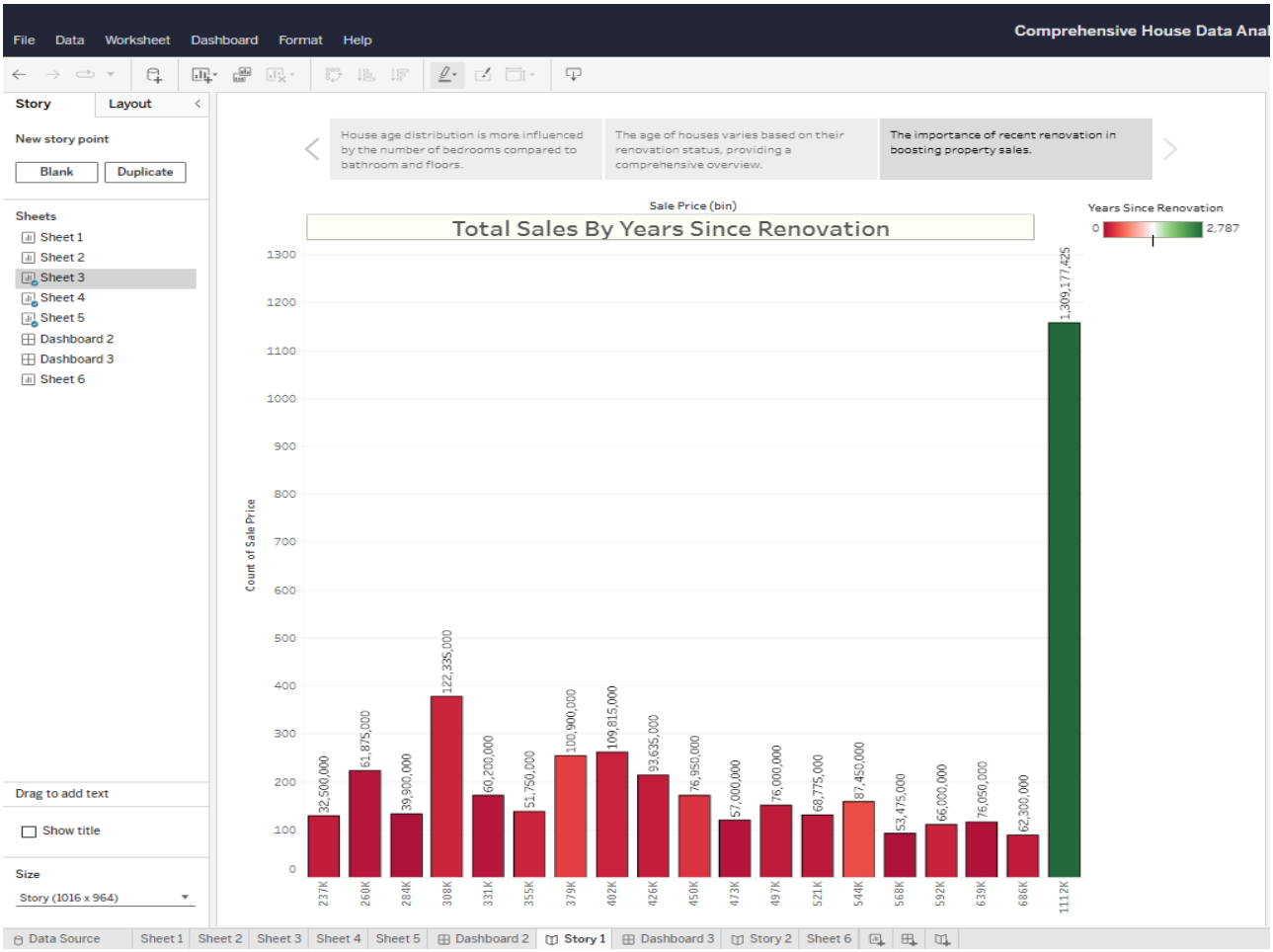
b) Here I have to add 3 sheets into one story so click on **Blank** .



c) Drag and Drop Sheets from Story panel into white Screen and add caption for each sheet.


File Data Worksheet Dashboard Format Help
Comprehensive House Data Analysis (Tableau Public)
Story
Layout
New story point
Blank
Duplicate
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Dashboard 3
Drag to add text
Show title
Size
Story (1016 x 964)
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovation in boosting property sales.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
50
51
59
64
71
Value
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Story 1
Dashboard 3
Story 2


File Data Worksheet Dashboard Format Help
Comprehensive House Data Analysis (Tableau Public)
Story
Layout
New story point
Blank
Duplicate
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Dashboard 3
Drag to add text
Show title
Size
Story (1016 x 964)
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovation in boosting property sales.
Distribution of House Age by Renovation Status
41
50
51
56
59
63
64
68
71
76
Age of House (in Y...
41
50
51
56
59
63
64
68
71
76
Ever Renovated Yes
166
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Story 1
Dashboard 3
Story 2

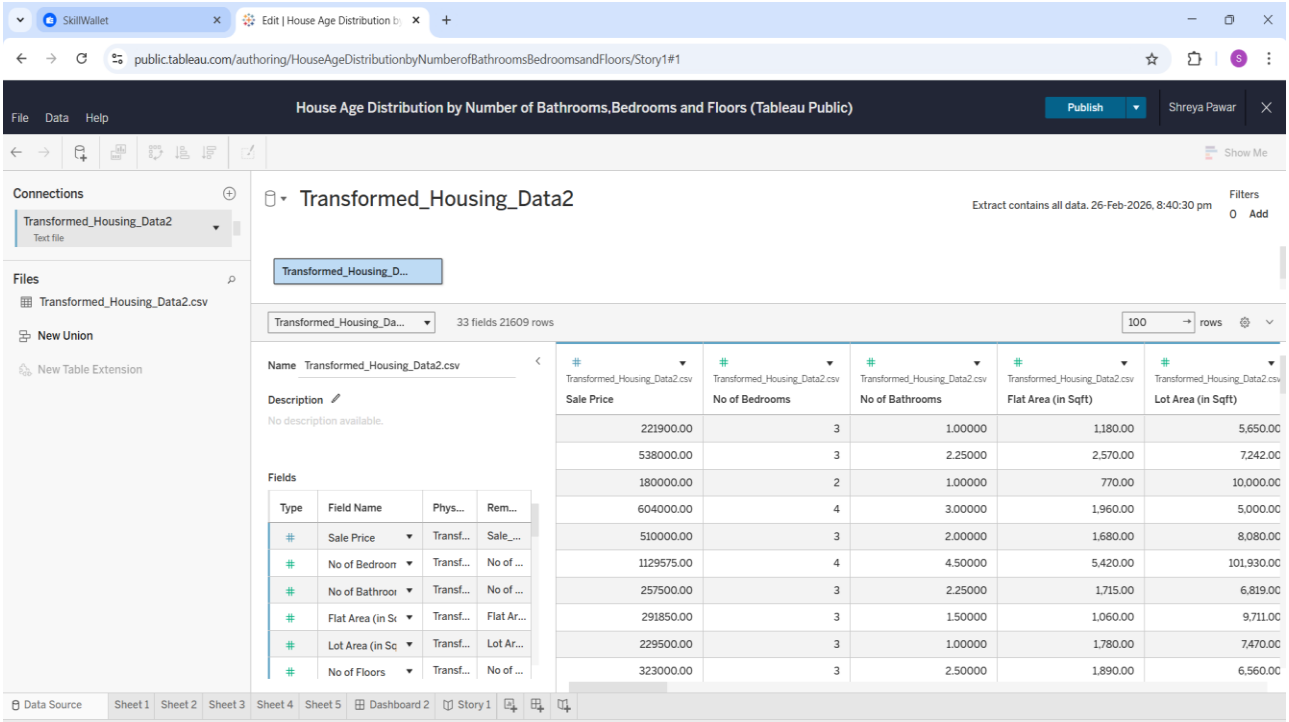


6) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

❖ Amount of Data Loaded

"Amount of Data Loaded" refers to the quantity or volume of data that has been imported, retrieved, or loaded into a system, software application, database, or any other data storage or processing environment. It's a measure of how much data has been successfully processed and made available for analysis, manipulation, or use within the system.

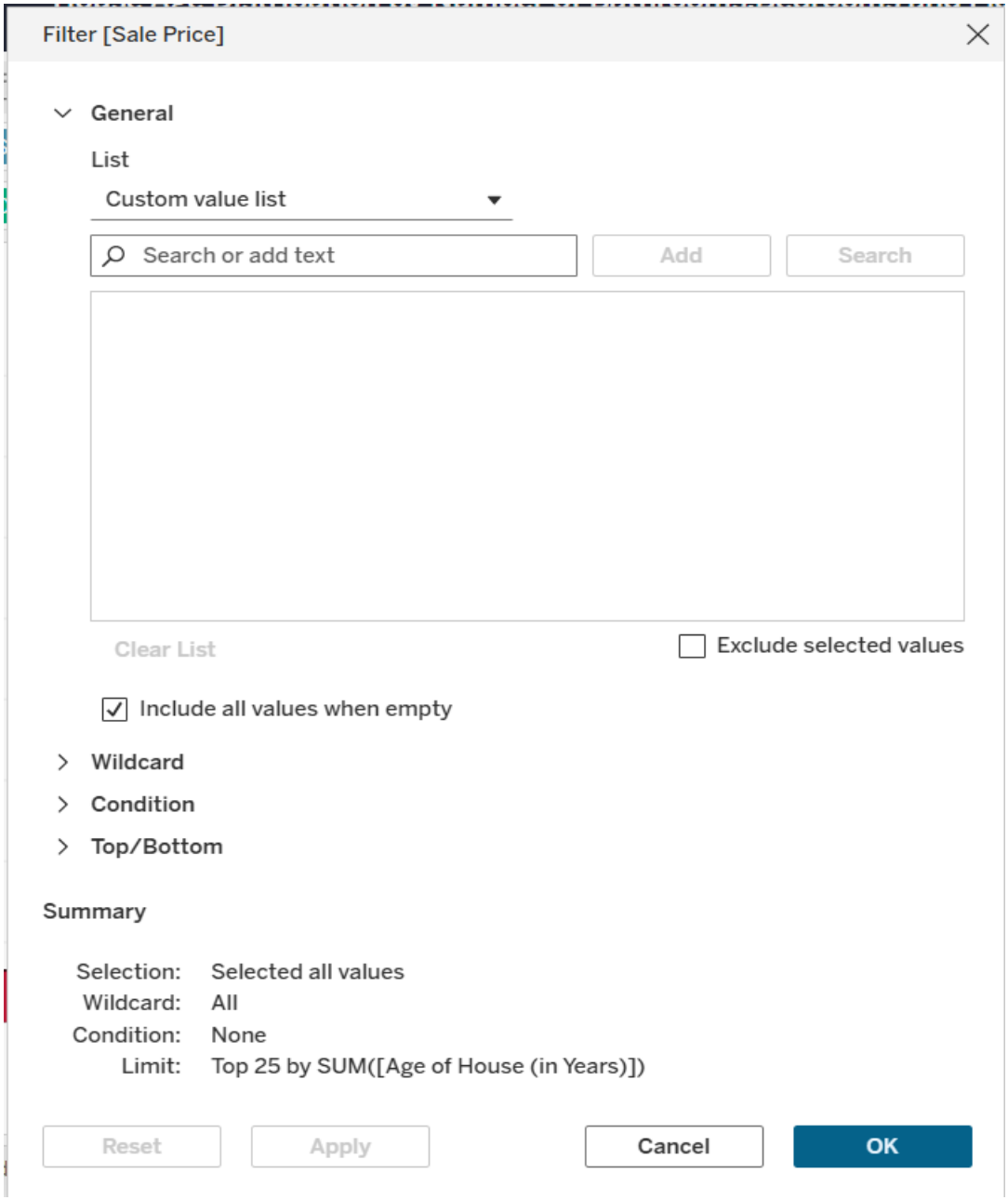


Utilization of Filters

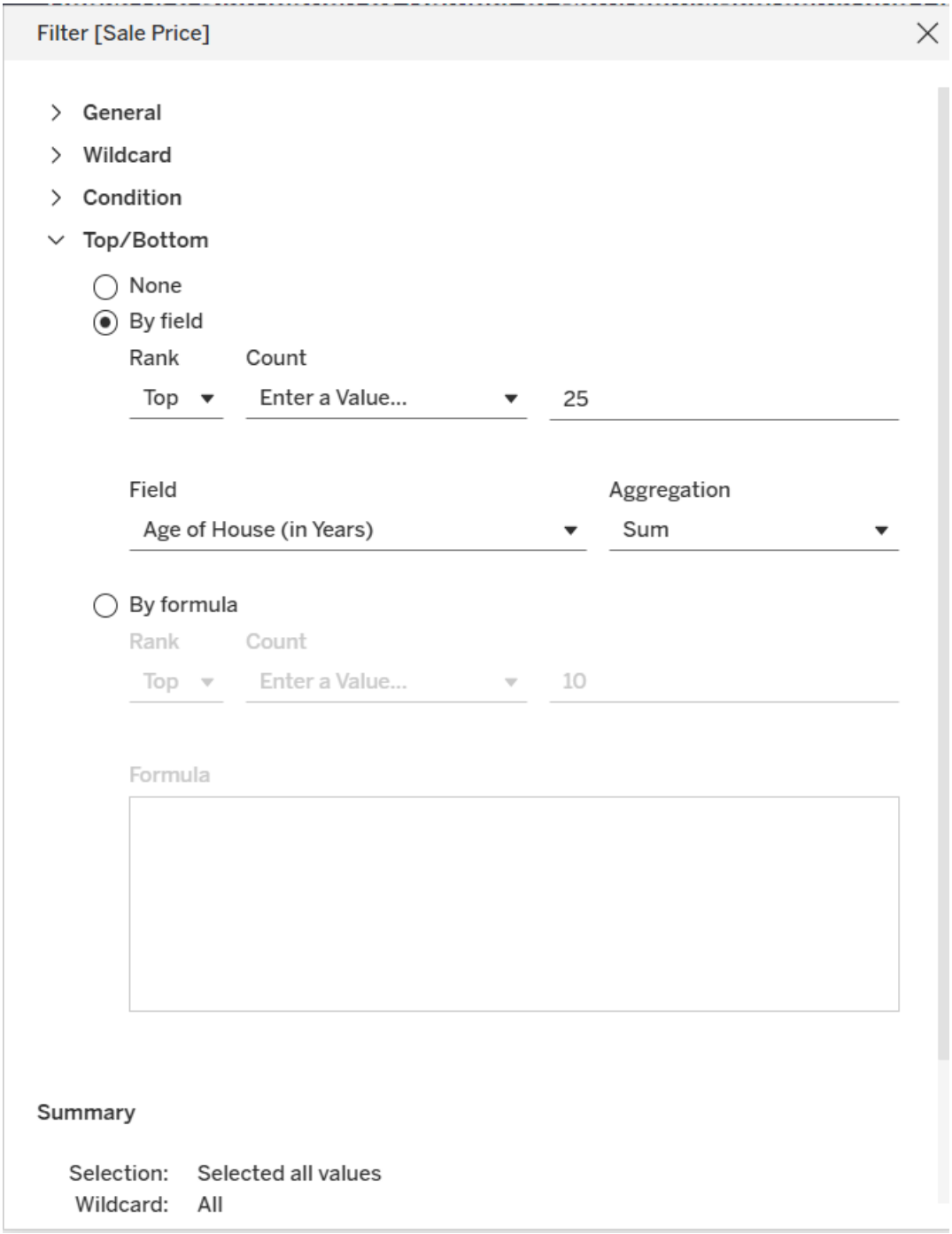
"Utilization of Filters" refers to the application or use of filters within a system, software application, or data processing pipeline to selectively extract, manipulate, or analyse data based on specified criteria or conditions.

1) Selected "Sales Price" as a Top N Filters.

a) Drage and drop "Sales Price" into Filters Filed.
b) Right click on "Sales Price" and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.

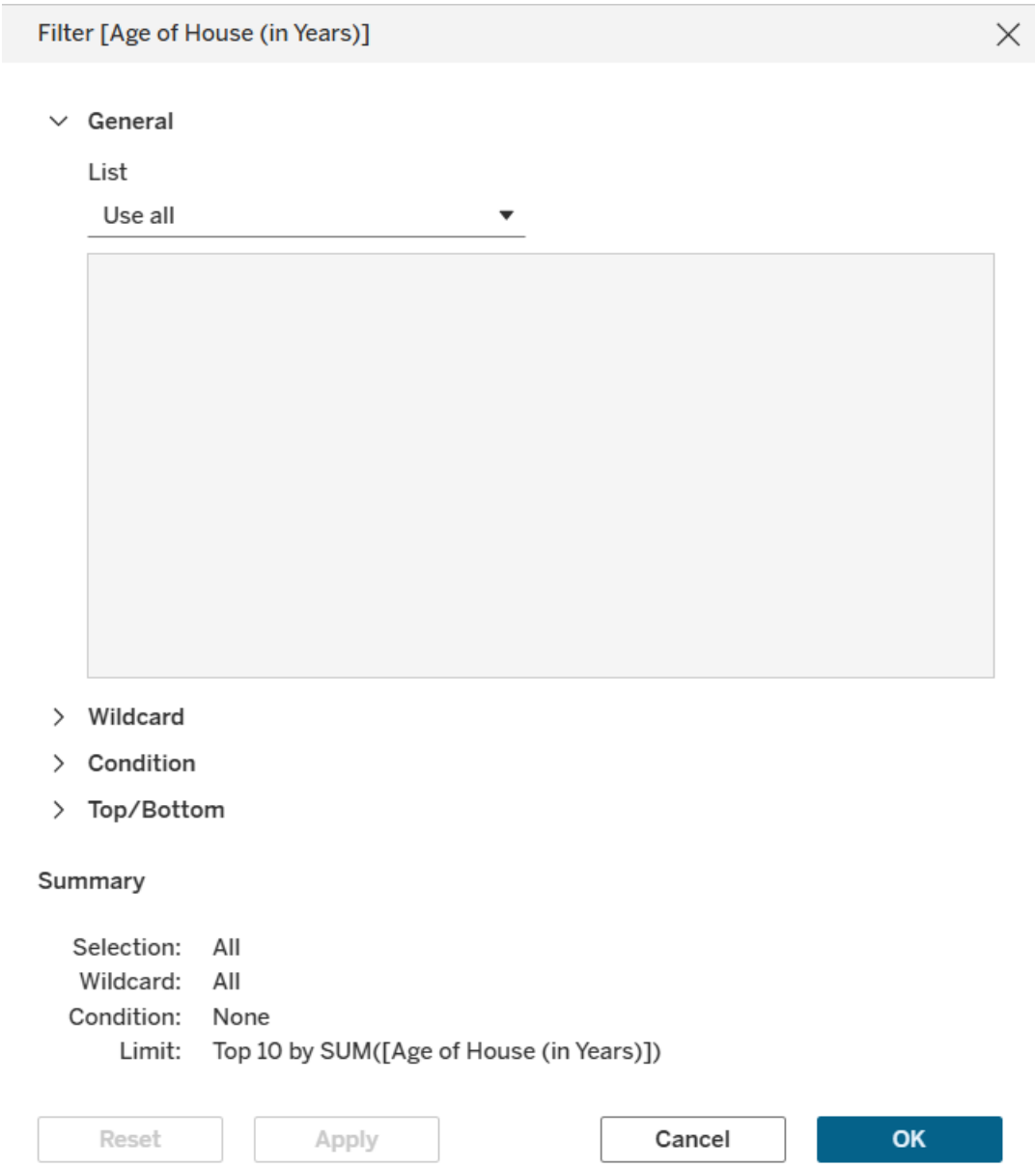


d) Select **By Field** and add Top 25 records and click OK.

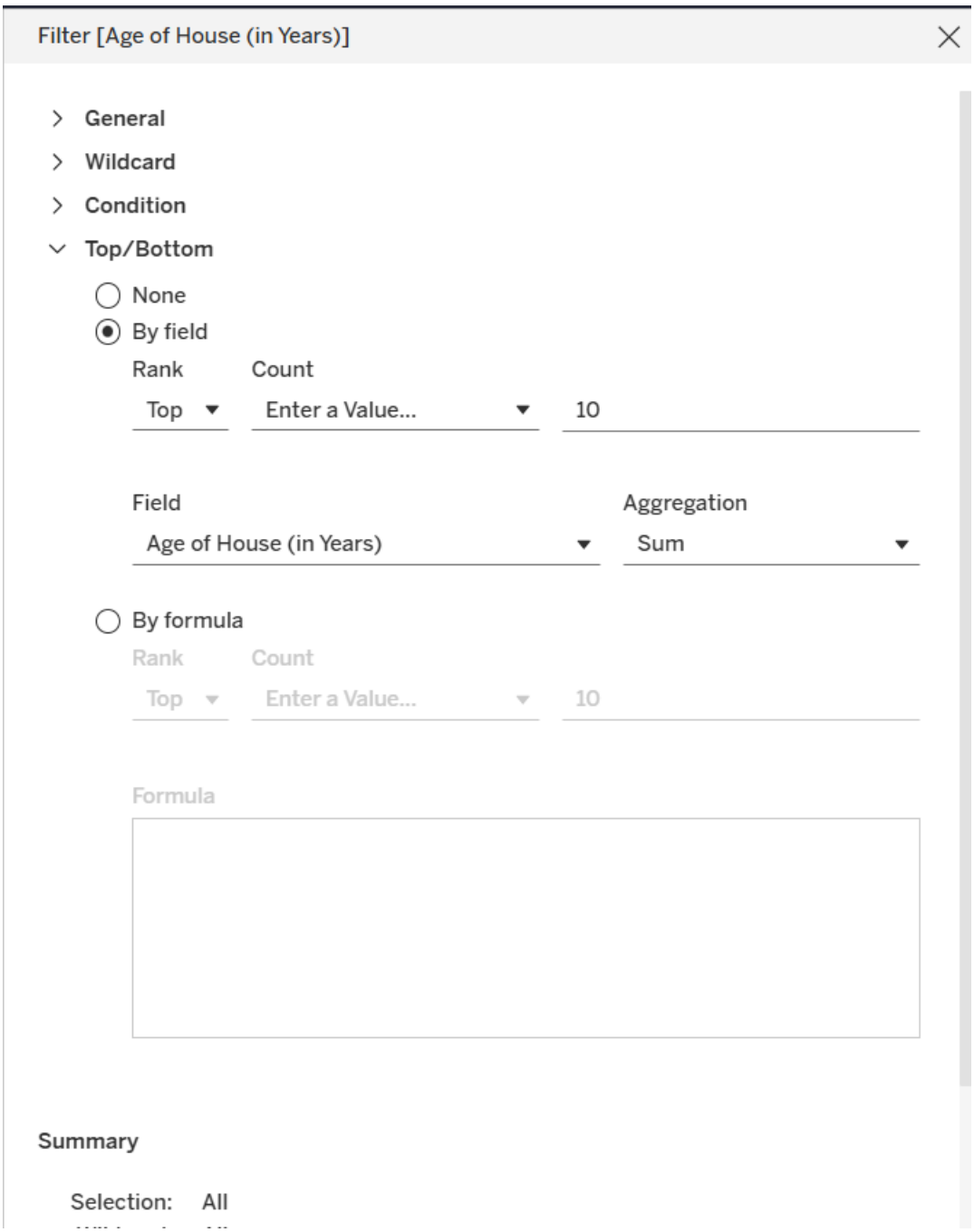


2) **Select "Age of House (in Year)" as a Top N records.**

a) Drage and drop “Age of House (in Year)” into Filters Filed.
b) Right click on “Sales Price” and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.



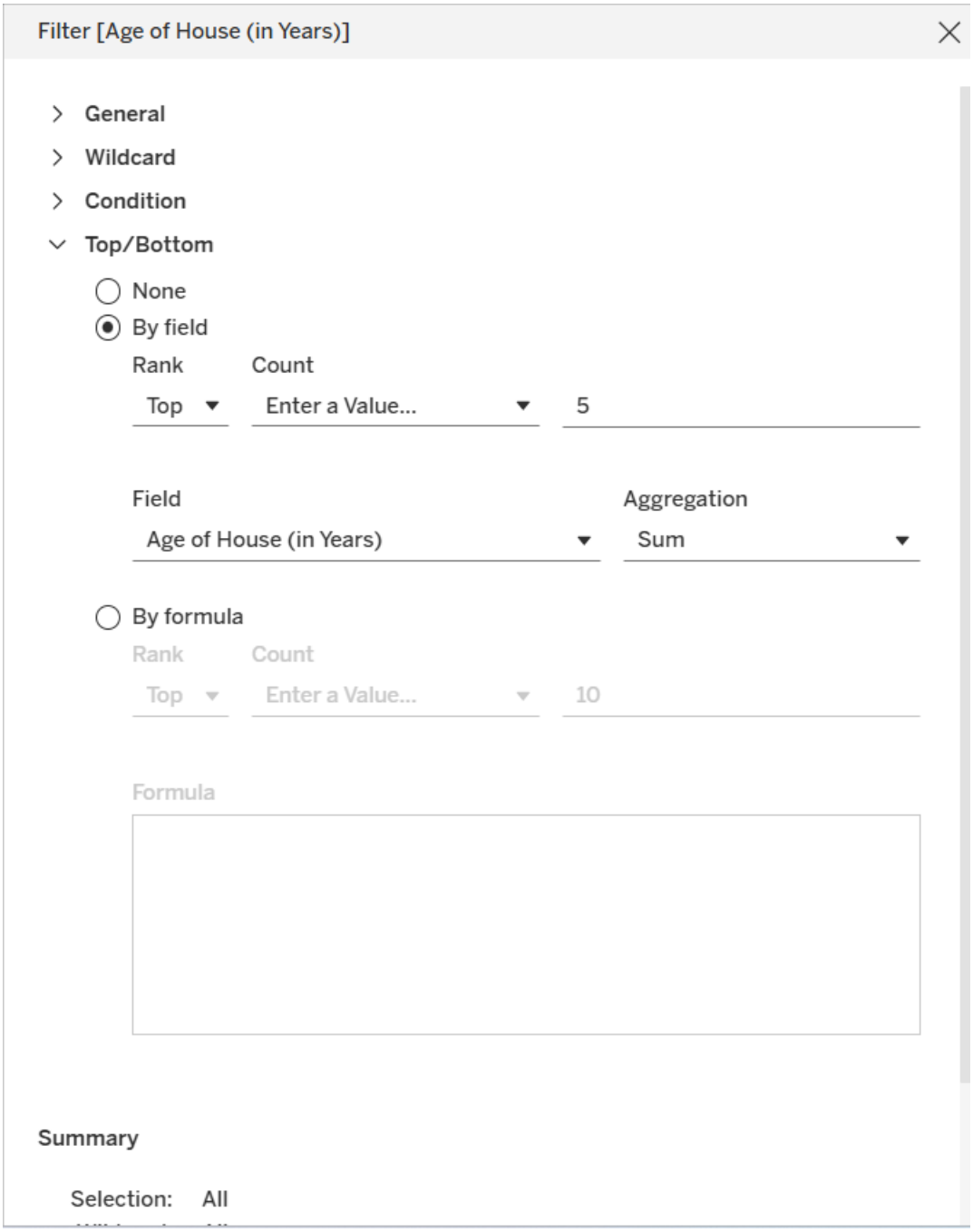
d) Select **By Field** and add Top 10 records and click OK.



3) **House Age Distribution by Top N Filters.**

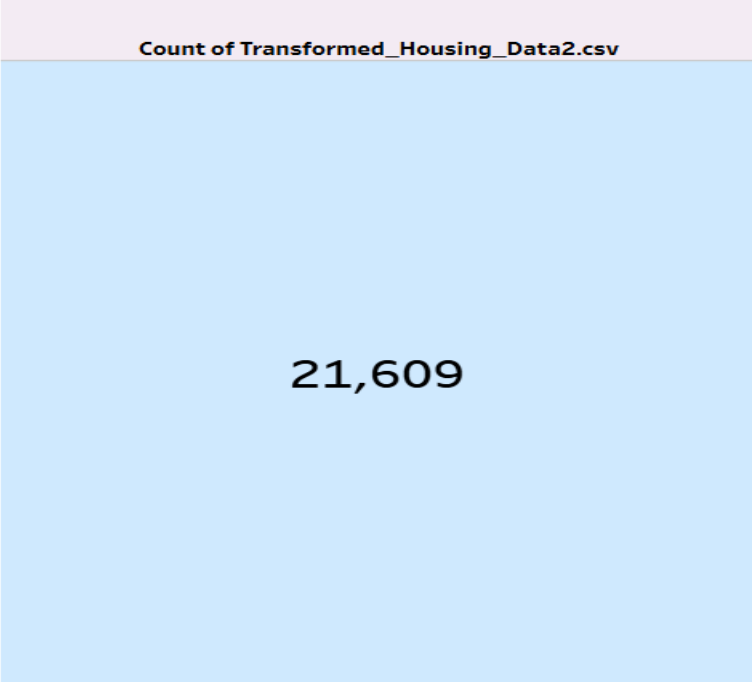
a) Drage and drop "Age of House (in Year)" into Filters Filed.
b) Right click on "Sales Price" and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.

d) Select **By Field** and add Top 5 records and click OK.



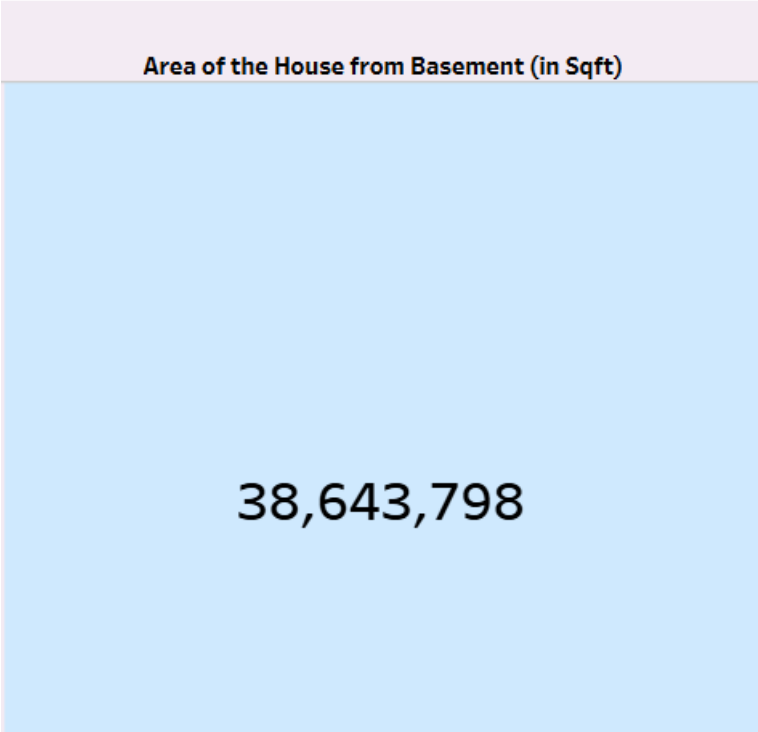
❖ No of Visualizations/ Graphs

1) Count of Transformed Housing Data

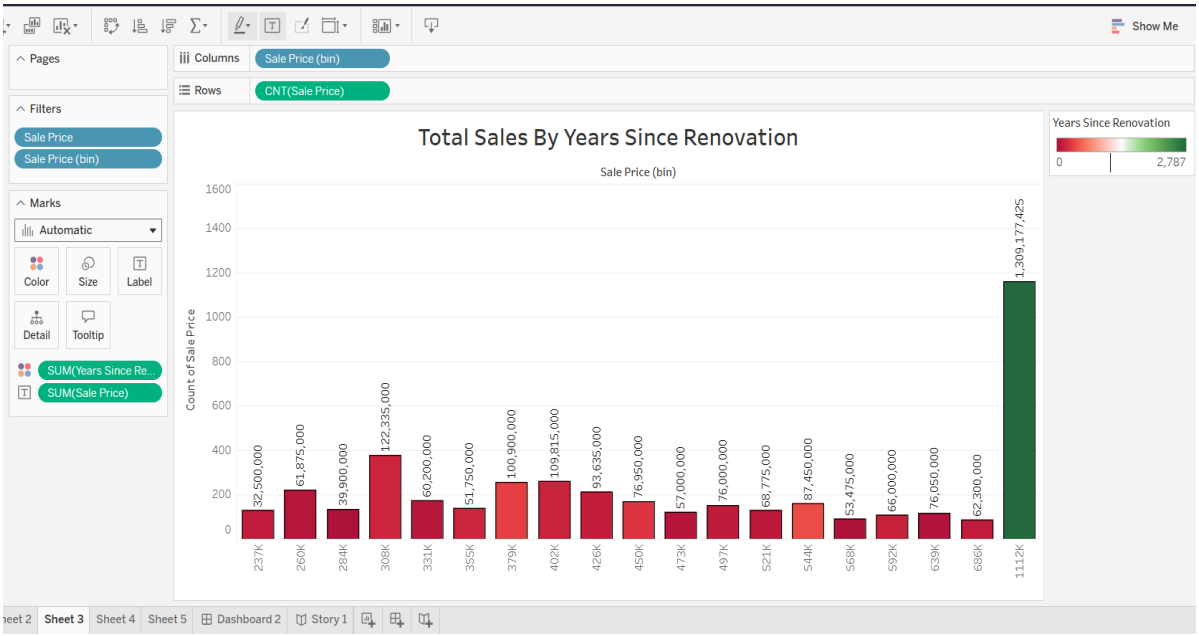


2) Average Sale Prices

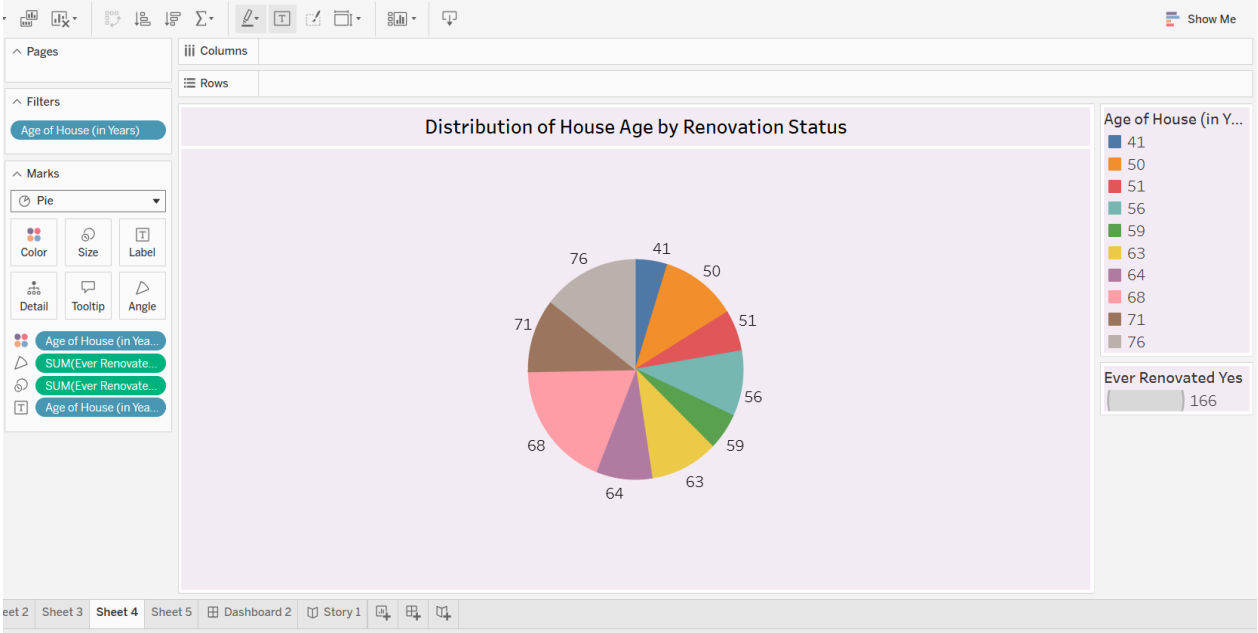
3) Area of House from Basement (in Sqft)



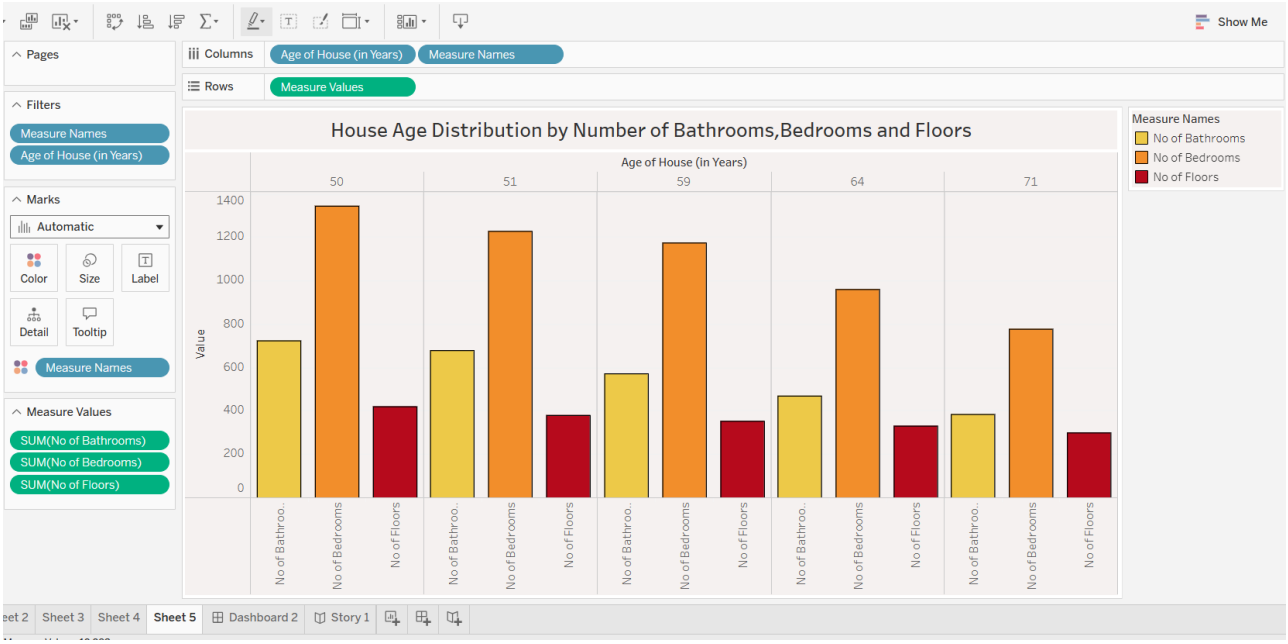
4) Total Sales by Years Since Renovation



5) Distribution of House Age by Renovation Status



6) House Age Distribution by Number of Bathrooms, Bedrooms, and Floors



7) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

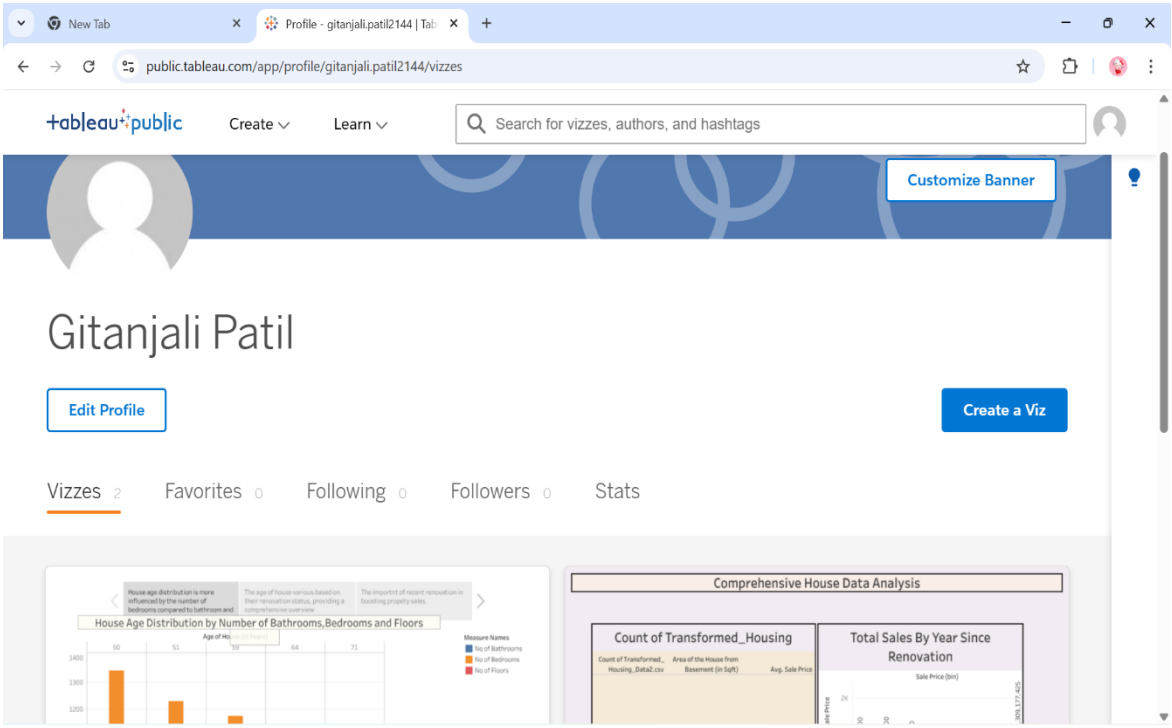
❖ Publishing

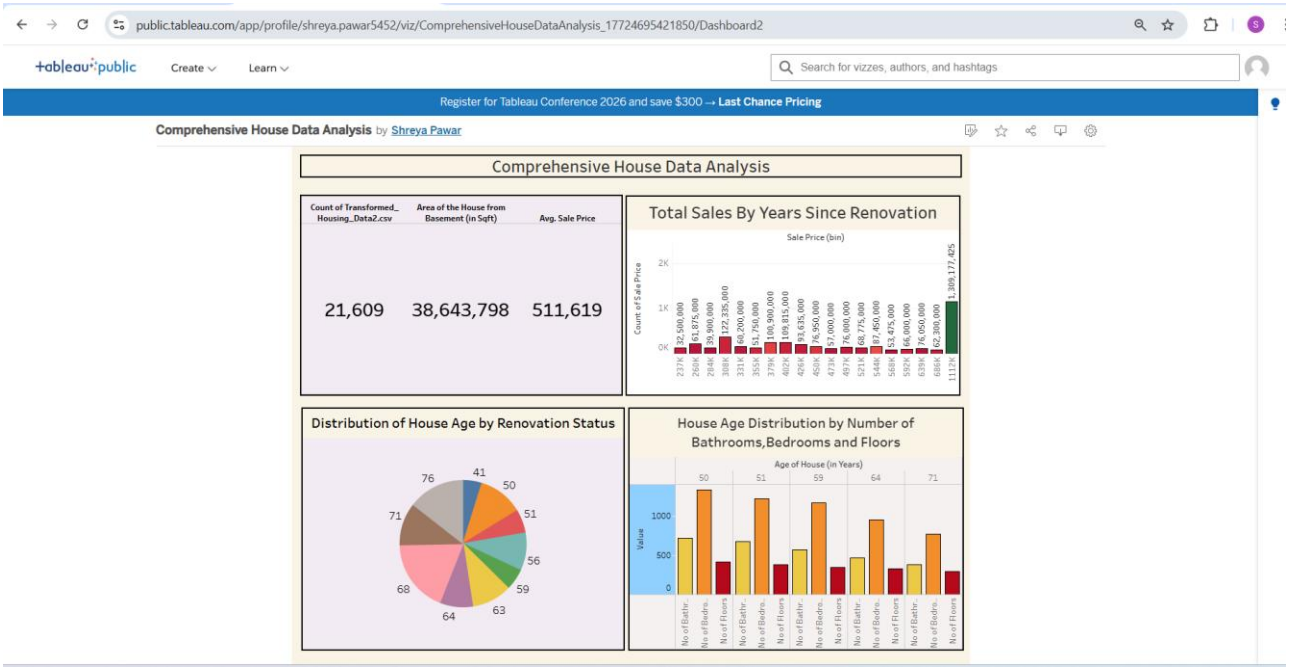
a) **Sign in to Tableau Public**

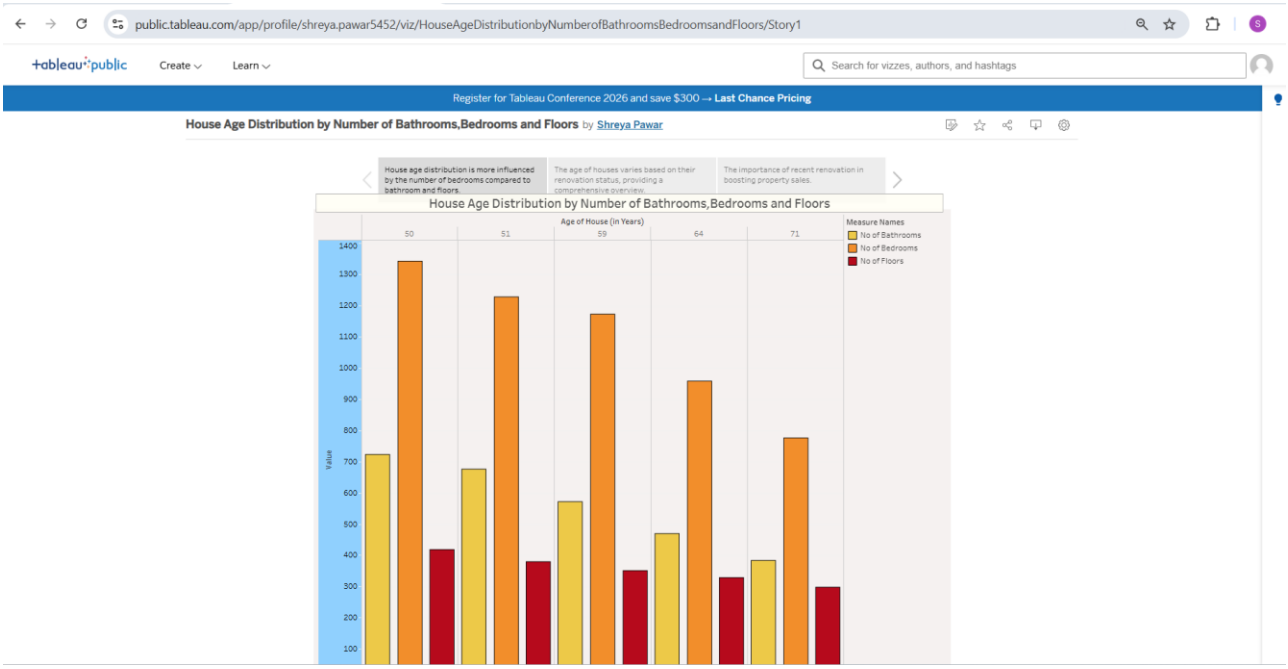
In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.



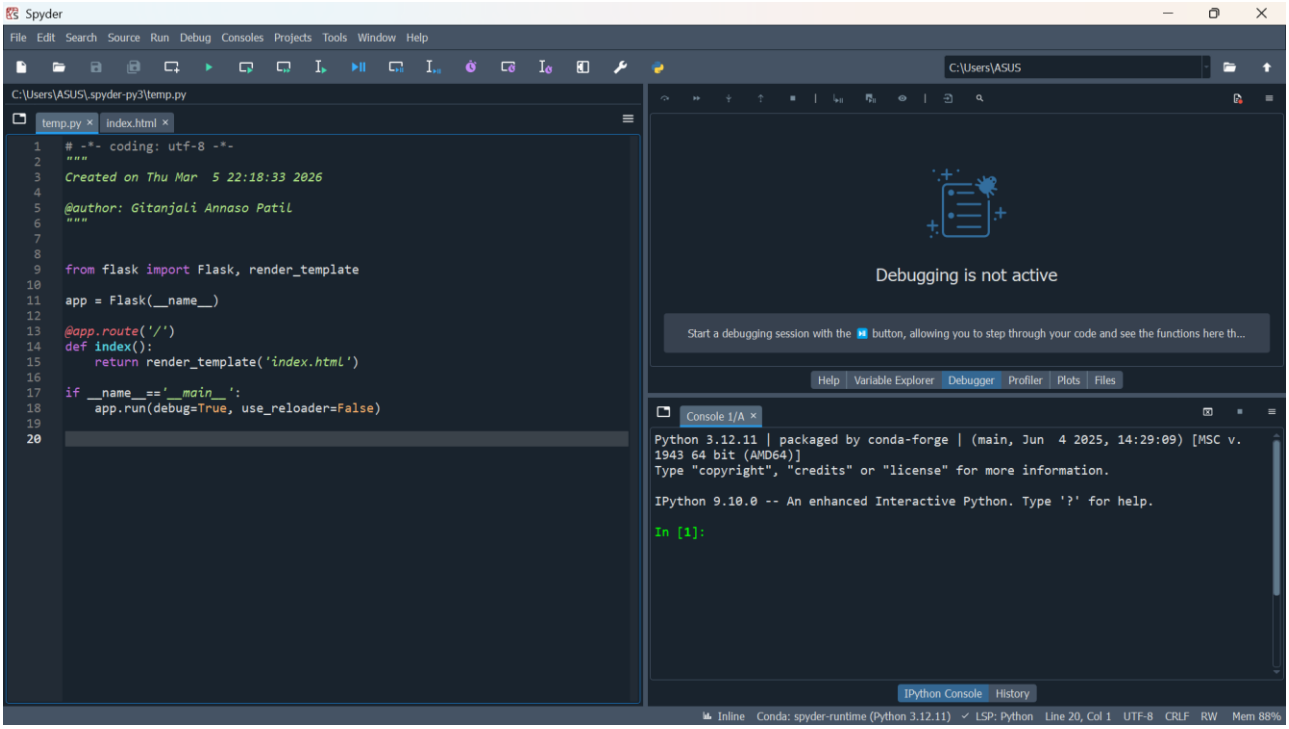

public.tableau.com/app/profile/shreya.pawar5452/viz/ComprehensiveHouseDataAnalysis_17724695421850/Dashboard2
tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Register for Tableau Conference 2026 and save $300 → Last Chance Pricing
Comprehensive House Data Analysis by Shreya Pawar
Comprehensive House Data Analysis
Count of Transformed_Housing_Data2.csv
Area of the House from Basement (in Sqft)
Avg. Sale Price
21,609
38,643,798
511,619
Total Sales By Years Since Renovation
Sale Price (bin)
Count of Sale Price
Distribution of House Age by Renovation Status
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Value


public.tableau.com/app/profile/shreya.pawar5452/viz/HouseAgeDistributionbyNumberofBathroomsBedroomsandFloors/Story1
tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Register for Tableau Conference 2026 and save $300 → Last Chance Pricing
House Age Distribution by Number of Bathrooms,Bedrooms and Floors by Shreya Pawar
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview...
The importance of recent renovation in boosting property sales.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Value

❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) temp.py

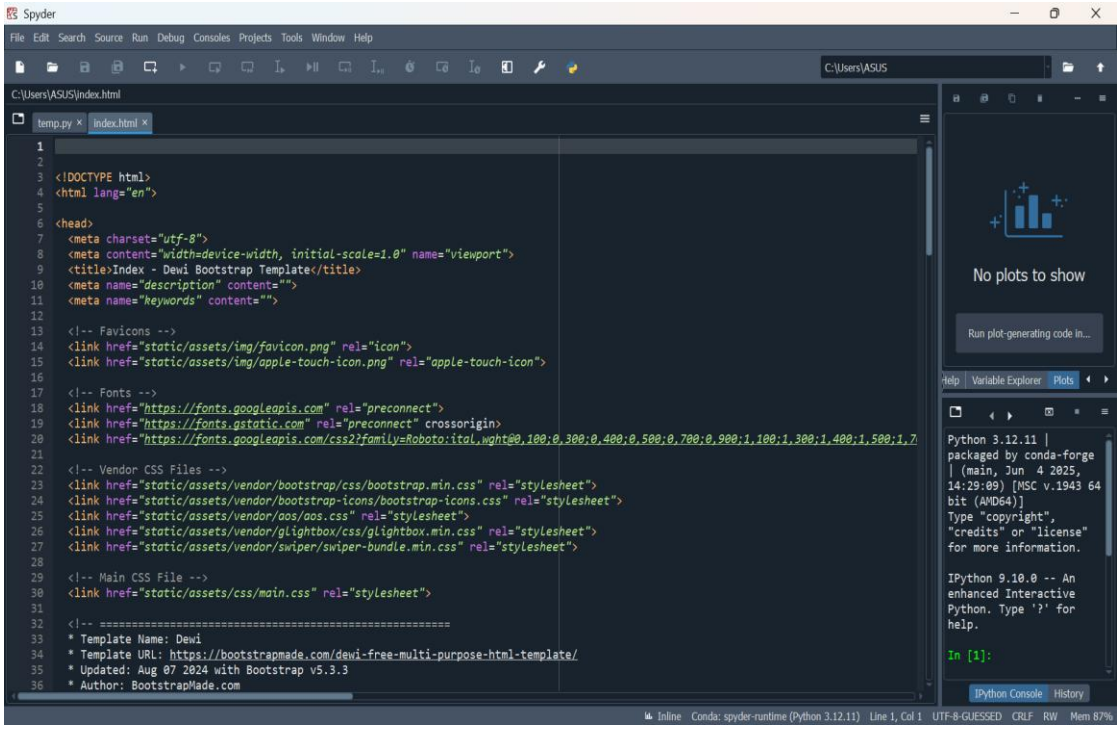


2) Adding Dashboard and Story on HTML page.

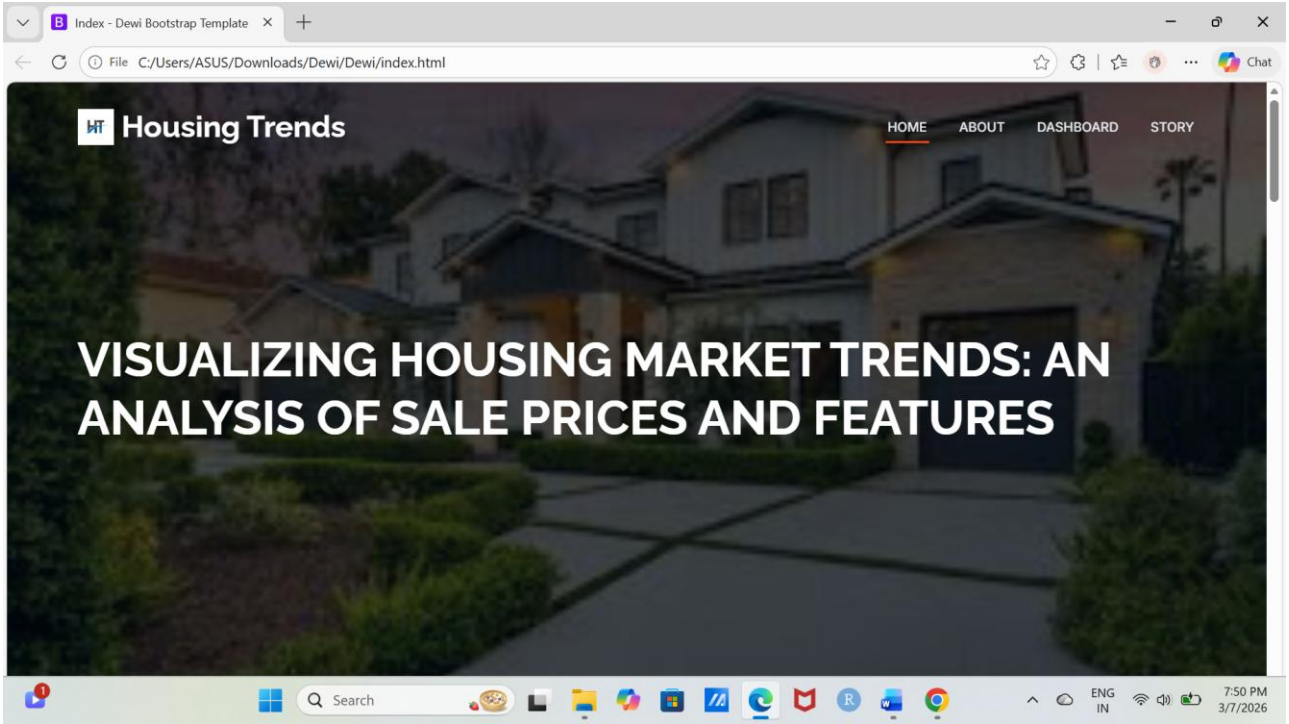
a) Open Tableau Public profile and open Dashboard/Story.

b) Click on share option and copy the embed code.

c) Paste in HTML page



d) Run the temp.py and copy the port number and paste it on Browser.


Index - Dewi Bootstrap Template
File C:/Users/ASUS/Downloads/Dewi/Dewi/index.html
Chat
Housing Trends
HOME
ABOUT
DASHBOARD
STORY
VISUALIZING HOUSING MARKET TRENDS: AN ANALYSIS OF SALE PRICES AND FEATURES
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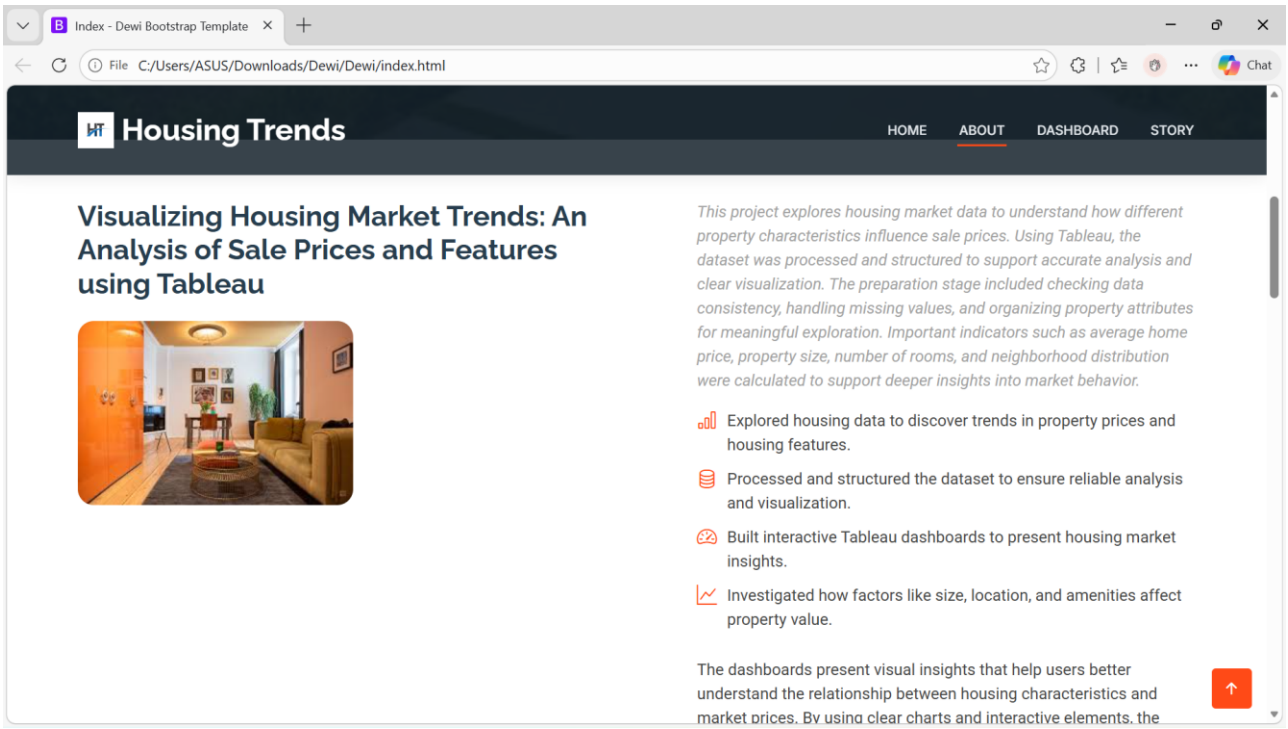
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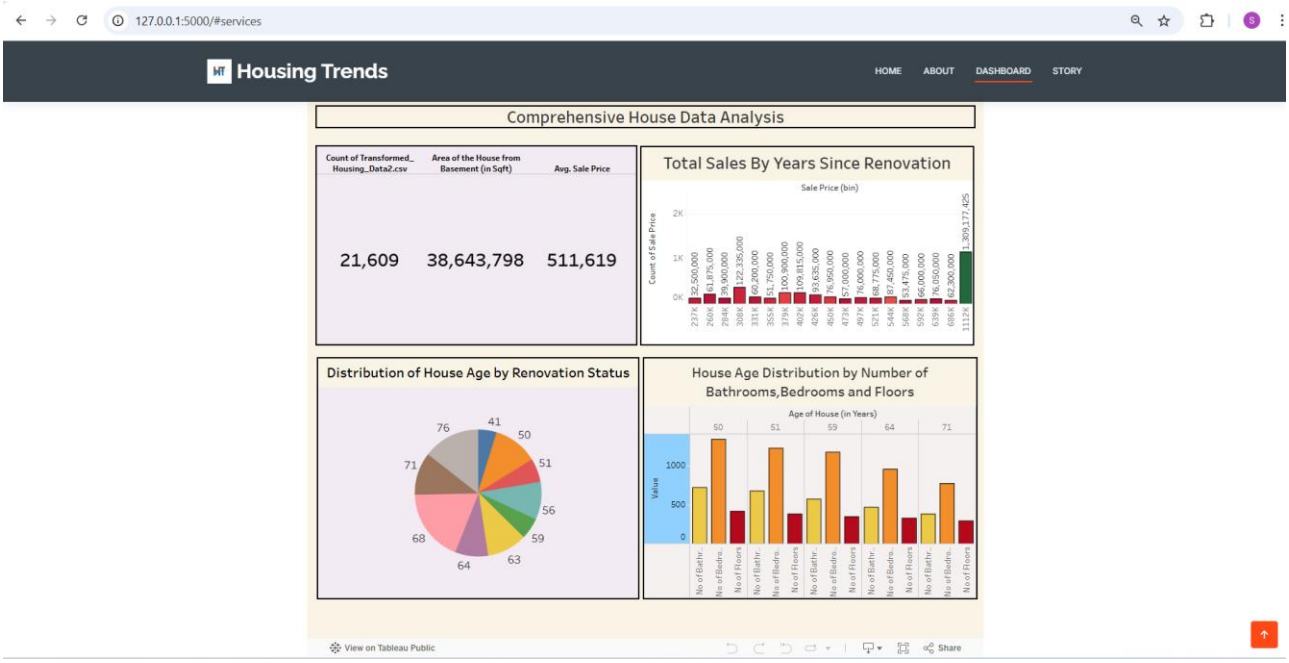
Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
This project explores housing market data to understand how different property characteristics influence sale prices. Using Tableau, the dataset was processed and structured to support accurate analysis and clear visualization. The preparation stage included checking data consistency, handling missing values, and organizing property attributes for meaningful exploration. Important indicators such as average home price, property size, number of rooms, and neighborhood distribution were calculated to support deeper insights into market behavior.
Explored housing data to discover trends in property prices and housing features.
Processed and structured the dataset to ensure reliable analysis and visualization.
Built interactive Tableau dashboards to present housing market insights.
Investigated how factors like size, location, and amenities affect property value.
The dashboards present visual insights that help users better understand the relationship between housing characteristics and market prices. By using clear charts and interactive elements, the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Comprehensive House Data Analysis
Count of Transformed_Housing_Data2.csv
Area of the House from Basement (in Sqft)
Avg. Sale Price
21,609
38,643,798
511,619
Total Sales By Years Since Renovation
Sale Price (bin)
Count of Sale Price
Distribution of House Age by Renovation Status
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Value
View on Tableau Public
Share

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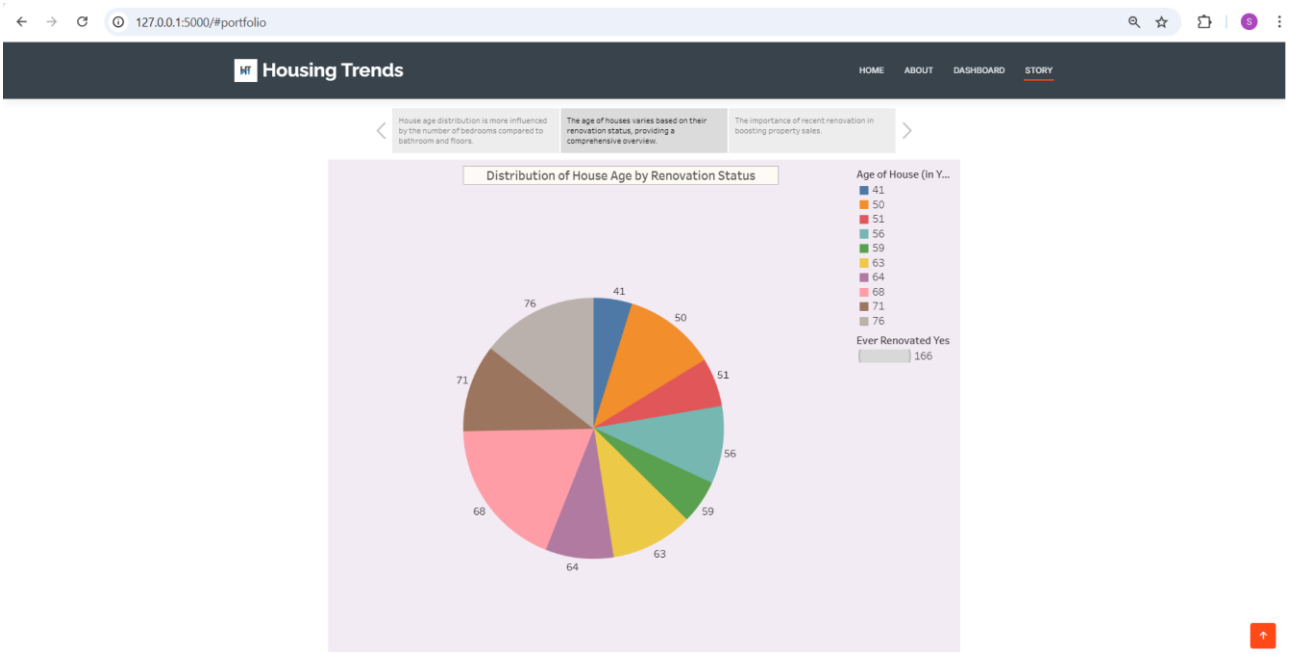
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview...
The importance of recent renovation in boosting property sales.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Value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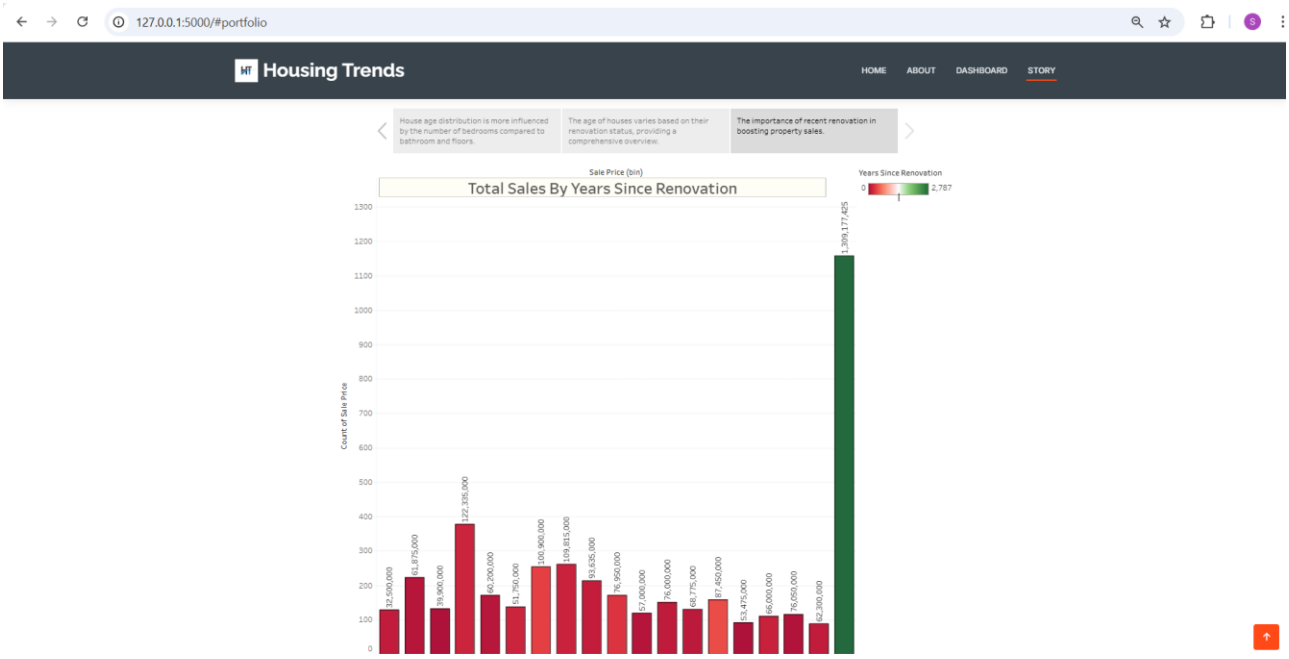
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovation in boosting property sales.
Distribution of House Age by Renovation Status
Age of House (in Y...
41
50
51
56
59
63
64
68
71
76
Ever Renovated Yes
166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Housing Trends
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House age distribution is more influenced by the number of bedrooms compared to bathroom and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovation in boosting property sales.
Sale Price (bin)
Total Sales By Years Since Renovation
Years Since Renovation
0
2,787
Count of Sale Price
1300
1200
1100
1000
900
800
700
600
500
400
300
200
100
0
32,500,000
61,875,000
39,900,000
122,335,000
60,200,000
51,750,000
100,900,000
109,815,000
93,635,000
76,950,000
57,000,000
76,000,000
68,775,000
87,450,000
53,475,000
66,000,000
76,050,000
62,300,000
1,399,177,425